

Task 1A. Initial Data Exploration

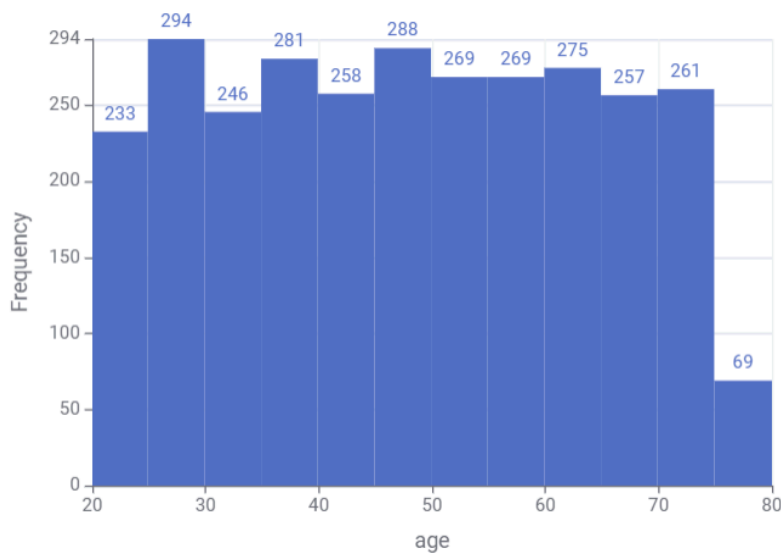
1A. 1 – Attribute Type Classification

Attribute	Attribute Type	Explanation
Source ID	Nominal	Unique identifier label. Arithmetic operations are meaningless.
Age	Ratio	True zero point (birth). 60 years is twice as old as 30 years.
Gender	Nominal	Categories (Male, Female, Other) with no inherent order.
Marital_status	Nominal	Categories (Single, Married, Divorced, Widowed) with no order.
Education_level	Ordinal	Has logical order (High School < Bachelor's < Master's < PhD) but unequal intervals.
Annual_income	Ratio	True zero (\$0). \$100,000 is twice \$50,000.
Monthly_income	Ratio	True zero (\$0). Meaningful ratios.
Employment_status	Nominal	Categories (Employed, Unemployed, Student, Retired) with no order.
Debt_to_income_ratio	Ratio	True zero (0%). 0.30 is three times 0.10.
Credit_score	Interval	Equal intervals between scores, but no true zero. A score of 0 does not mean "no creditworthiness."
loan_amount	Ratio	True zero (\$0). Meaningful ratios.
Loan_purpose	Nominal	Categories (Car, Business, Medical, etc.) with no order.
Interest_rate	Ratio	True zero (0%). 10% is twice 5%.
Loan_term	Ratio	True zero (0 months). 60 months is twice 30 months.
Installment	Ratio	True zero (\$0). Meaningful ratios.
Num_of_open_accounts	Ratio	Count data with true zero (0 accounts). 4 is twice 2.
Total_credit_limit	Ratio	True zero (\$0). Meaningful ratios.
Current_balance	Ratio	True zero (\$0). Meaningful ratios.
Delinquency_history	Ratio	Count data with true zero.
Num_of_delinquencies	Ratio	Count data with true zero.
Loan_paid_back	Nominal	Binary category (1=Yes, 0=No). Numbers are labels, not quantities.

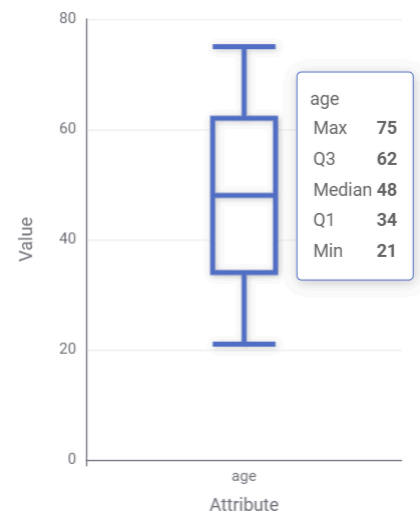
1A. 2 – Statistics Summarisation

1. Source ID: does not carry any meaningful information, so no statistics available.
2. Age: Ratio datatype, ranging from 21 to 75 years. Distribution is approximately uniform across all age groups (69 entries for 70 - 75 years of age). No single age group dominates, and no outliers are present. Histogram with equi-width bin of 5 years.

Age Frequency, bin = 5 years

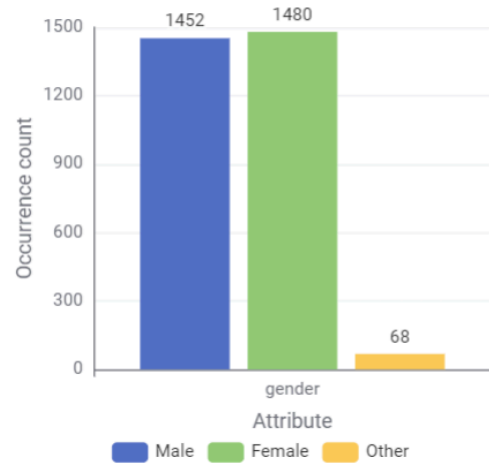


Age Box Plot



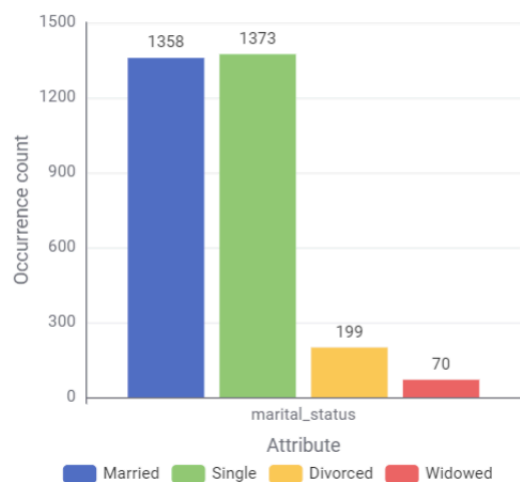
- Gender: Nominal. Balanced Female (49.3%) and Male (48.4%) distribution with a small “Other” (2.3%) category. No significant gender imbalance affecting analysis.

Gender Bar Chart



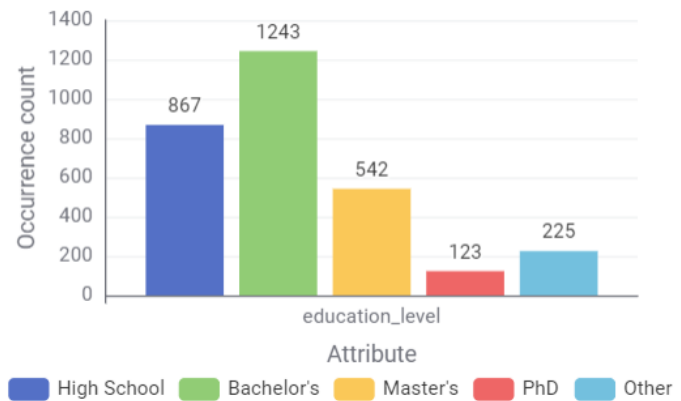
- Marital status: Nominal. The majority of borrowers are either Single (n=1,373, 45.8%) or Married (n=1,358, 45.3%), together accounting for 91.1% of the sample. Divorced borrowers represent 6.6% (n=199), while Widowed borrowers are the smallest group at 2.3% (n=70).

Marital Status Bar Chart



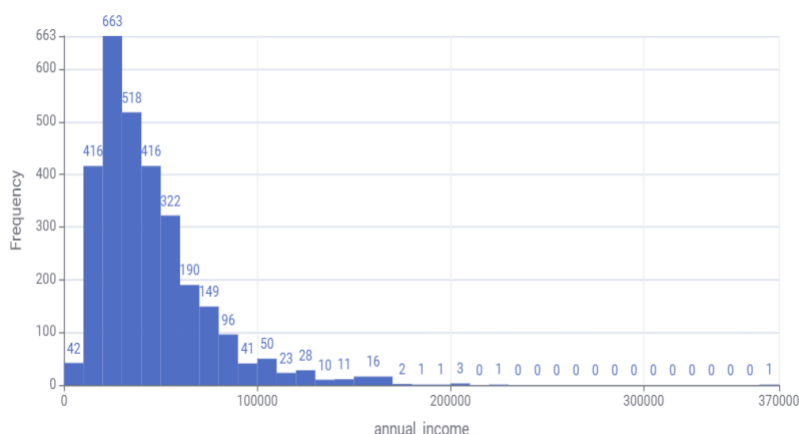
5. Education level: Ordinal with a natural order from High School to PhD. The most common level is Bachelor's degree (41.4%), followed by High School (28.9%). There is a monotonic decline in frequency as education level increases beyond Bachelor's: Master's (18.1%), PhD (4.1%). The 'Other' category (7.5%) captures responses that do not fit the standard classification.

Education Level Bar Chart

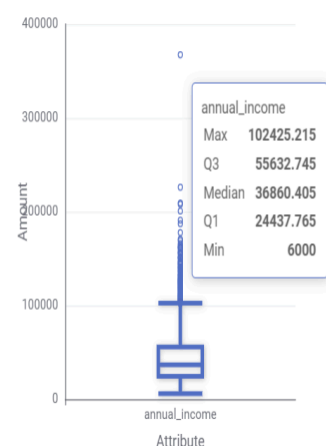


6. Annual income: Ratio datatype, from \$6,000 to \$367,316.11. Histogram: equi-width bins of \$10,000. Distribution: Right-skewed (most borrowers have lower incomes, with a long tail of high earners). A significant number of outliers exist above \$102,425 up to \$367,316, outside of the box plot whiskers. Relationship with Monthly Income: Monthly income = Annual income ÷ 12. Therefore, all statistics scale proportionally (mean, median, quartiles divided by 12). The histogram shape remains identical when bin width is scaled to \$833.33 (\$10,000 ÷ 12).

Annual Income Frequency, bin width = 10,000

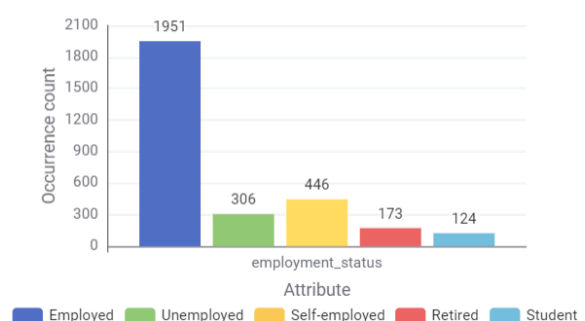


Annual Income Box Plot



7. Employment status: Nominal. The majority of borrowers are Employed (65%, n=1951), followed by Self-employed (14.9%, n=446).

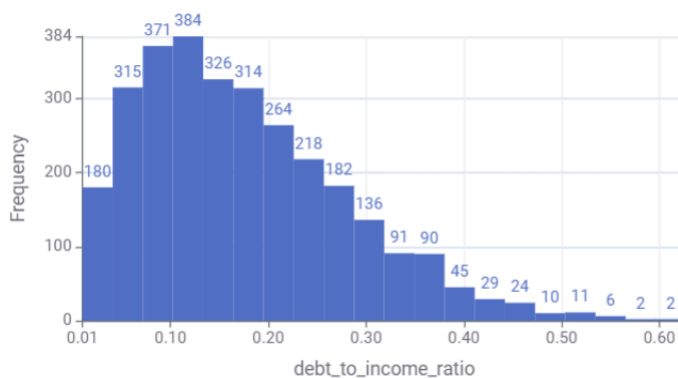
Employment Bar Chart



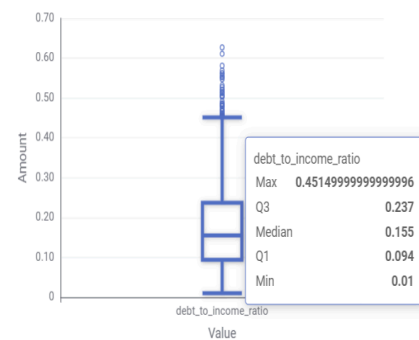
Unemployed, Retired, and Student represent smaller segments of the population."

8. Debt to income ratio: Ratio, range from 0.01 to 0.627. Equi-width binned histogram with 20 bins for frequency visualisation. Distribution: Right-skewed (most borrowers have low debt-to-income, fewer have high debt-to-income). Outliers exist above 0.451 up to 0.627. Key insight: The median DTI is 0.155, meaning half of borrowers have debt payments consuming less than 15.5% of their income. The interquartile range ($Q3 - Q1 = 0.143$) indicates moderate spread. The right tail extends to 0.627, indicating a minority of borrowers with high debt burdens.

Debt-to-income-ratio

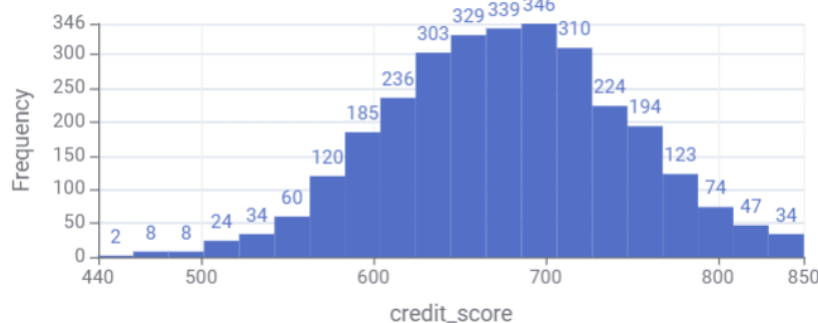


Debt-to-income-ratio Box Plot

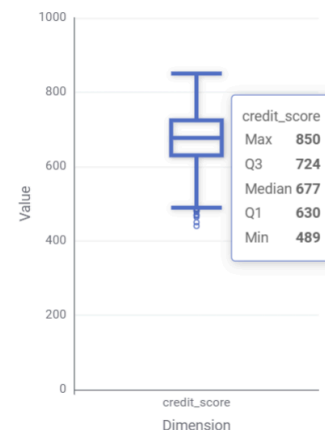


9. Credit score: Interval, from 440 to 850. 20 equi-width bins. Histogram: 20 equi-width bins (bin width = 20.5 points). Distribution: Approximately normal (bell-shaped), peaking in the 645-706 range. Box plot statistics: Min = 489, Q1 = 630, Median = 677, Q3 = 724, Max = 850. The interquartile range ($IQR = 94$) indicates moderate spread. The median of 677 falls slightly below the center of the range (645), suggesting a very slight left skew (more borrowers have scores below the midpoint). No extreme outliers are present, as the min (489) and max (850) fall within expected boundaries.

Credit Score

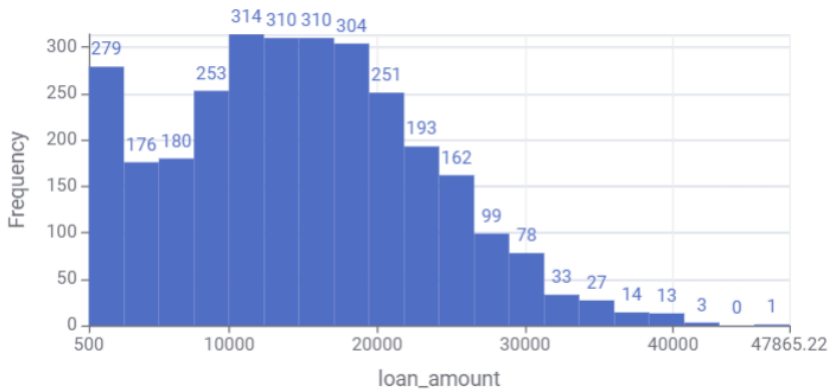


Credit Score

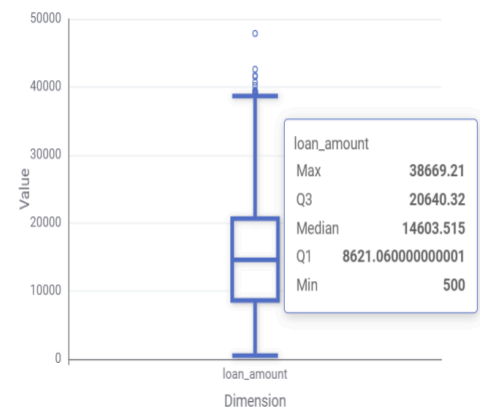


10. Loan amount: Ratio, range \$500 - \$47,865.22. 20 equi-width bins. Distribution: Right-skewed (most loans are small to medium, with a long tail of larger loans). The box plot upper whisker ends at \$38,669 ($Q3 + 1.5 \times IQR$). Values above this threshold, up to the true maximum of \$47,865, are outliers.

Loan Amount



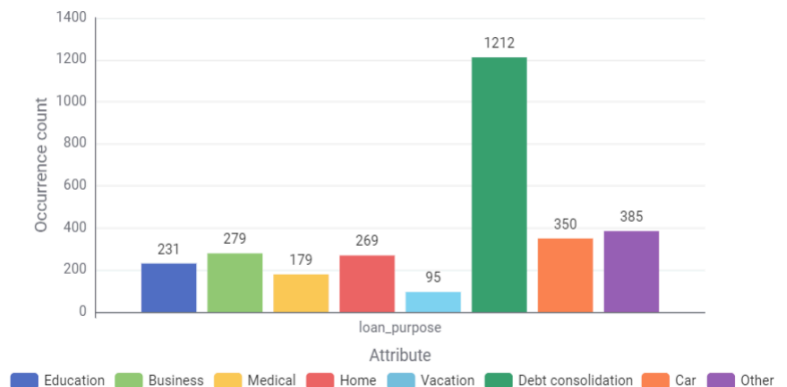
Loan Amount



11. Loan purpose: Nominal.

Distribution: Highly concentrated. debt consolidation dominates, while Vacation is the smallest category. Key insight: The predominance of debt consolidation suggests that most borrowers in this dataset are seeking loans to refinance or combine existing debts, rather than for new purchases or investments.

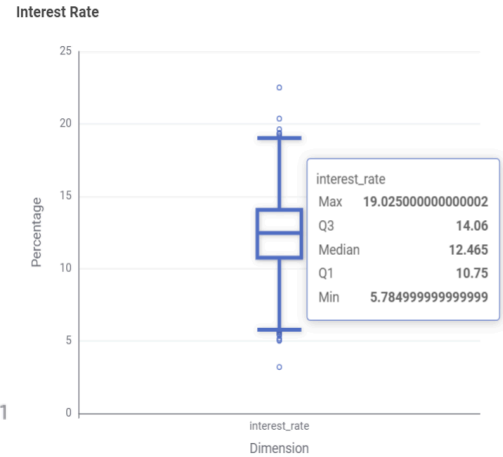
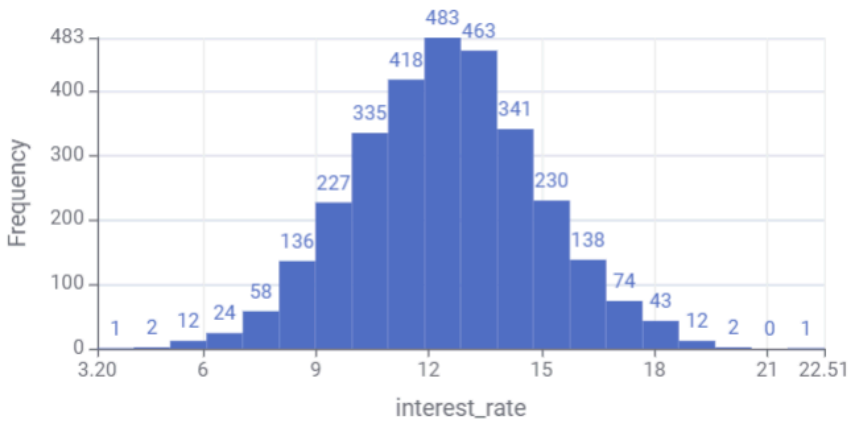
Loan Purpose



12. Interest rate: Ratio, range 3.2 - 22.51%. 20 equi-width bins.

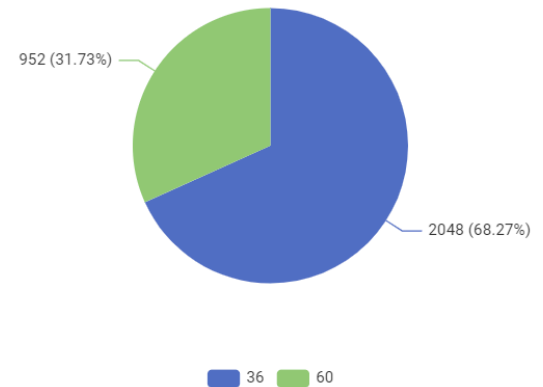
Distribution: Approximately normal (bell-shaped), peaking in the 12-13% range. The interest rate distribution is remarkably symmetric around 12.47%, with 50% of borrowers receiving rates between 10.75% and 14.06%. The presence of a small number of high-rate outliers (up to 22.51%) suggests a minority of borrowers with higher risk profiles.

Interest Rate



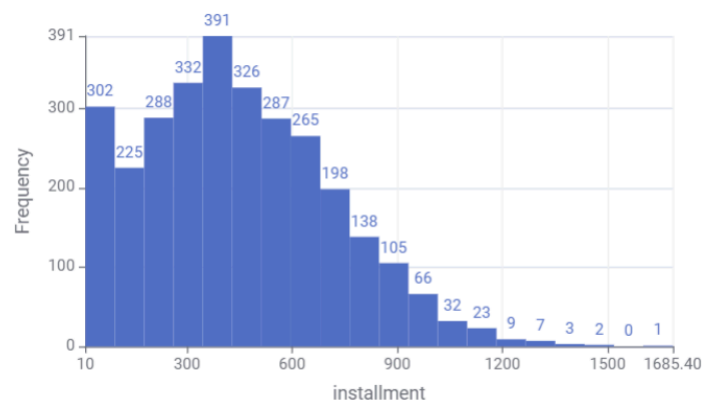
13. Loan term: Ratio. Distribution: Bimodal (only two possible values), with 60 months being more common. Distribution of loan terms. The 60-month term is more than twice as common as the 36-month term, accounting for over two-thirds of all loans. Key insight: The preference for 60-month loans suggests that borrowers prioritize lower monthly payments over paying less total interest. This is consistent with the debt consolidation trend observed in Loan Purpose, where borrowers may be seeking to minimize monthly obligations.

Loan Term (months)



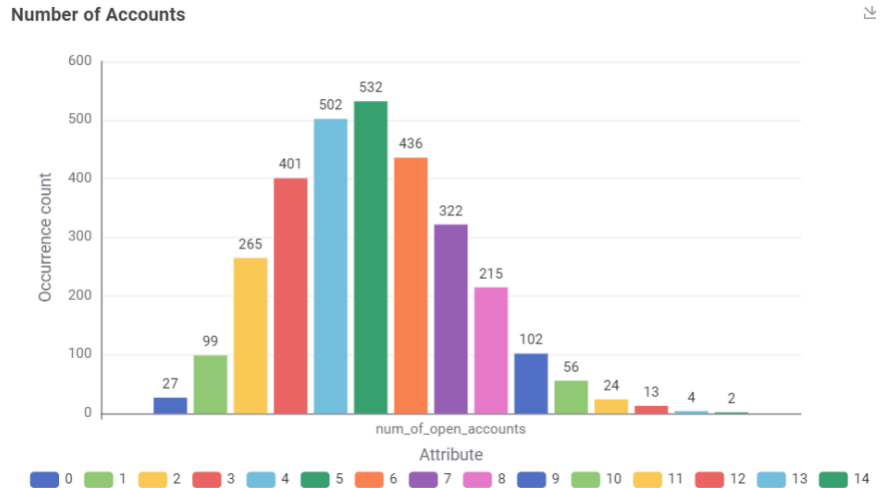
14. Installment: ratio, range 10 - 1685.4. 20 equi-width bins. Distribution: Right-skewed (most installments are low to moderate, fewer high installments). Key insight: Most borrowers have monthly installments below \$600 (bins 1-7 account for 2,257 borrowers, or 75.2% of the sample). Only 1.6% of borrowers (n=48) have installments above \$1,200.

Installment

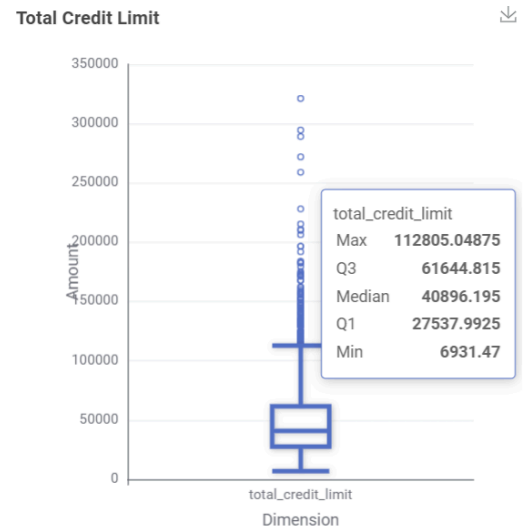
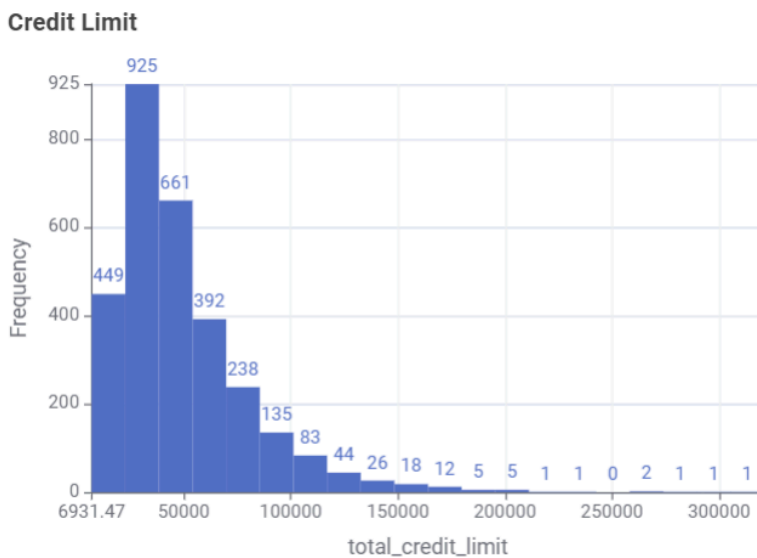


The near-empty bins at the high end (bins 19-20 contain only 1 borrower combined) indicate that very high monthly payments are rare, which is consistent with the typical loan amounts and interest rates observed.

15. Number of open accounts: Ratio, range 0 - 14. Distribution: Approximately normal (bell-shaped), peaking at 4-5 accounts. The median of 5 accounts is a typical profile for an active credit user.



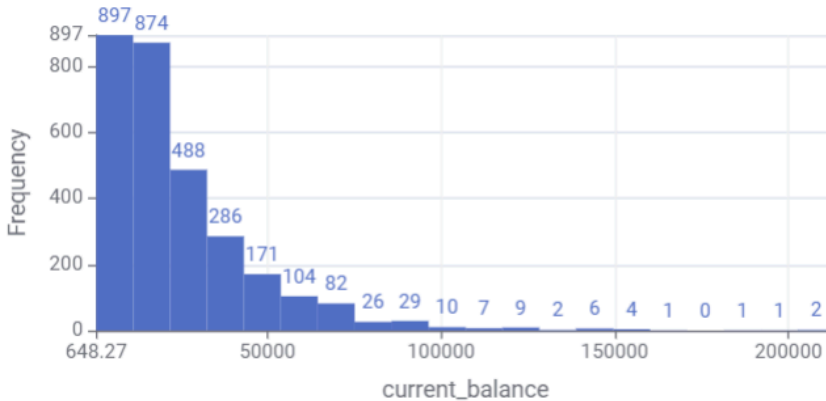
16. Total credit limit: Ratio, range: \$6931.47 - \$321097.22. 20 equi-width bins. Distribution: Right-skewed (most borrowers have lower credit limits, with a long tail of high limits). The highest concentration of borrowers (925, or 30.8%) falls within the lowest bin (\$6,931 - \$22,639). Frequencies decline steadily across bins, with only 12 borrowers (0.4%) having total credit limits above \$195,431. Key insight: The long tail of high credit limits suggests a small subset of borrowers with significantly greater access to credit, potentially representing higher-income individuals or those with established credit histories.



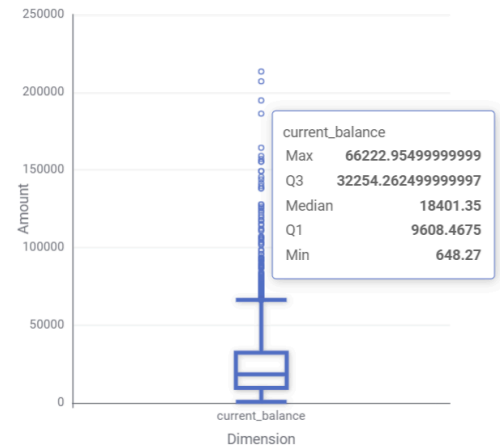
17. Current balance: Ratio, range \$648.27 - \$213,327.98. 20 equi-width bins. Distribution: Bimodal or irregular (peaks at both low and medium balances, with a long tail). Two distinct peaks in the distribution: A large peak in the lowest balance range (\$648 - \$50,000) with approximately 897 borrowers. A second peak in the

medium balance range (\$50,000 - \$100,000) with approximately 884 borrowers. Key insight: The bimodal distribution suggests two distinct borrower segments: Low-balance borrowers (median ~\$18K) — likely those who pay down balances regularly Medium-balance borrowers (peak around \$50-100K) — possibly those carrying significant debt The presence of high-balance outliers (up to \$213K) may represent a small subset of borrowers with either very high credit limits or significant financial distress.

Current Balance

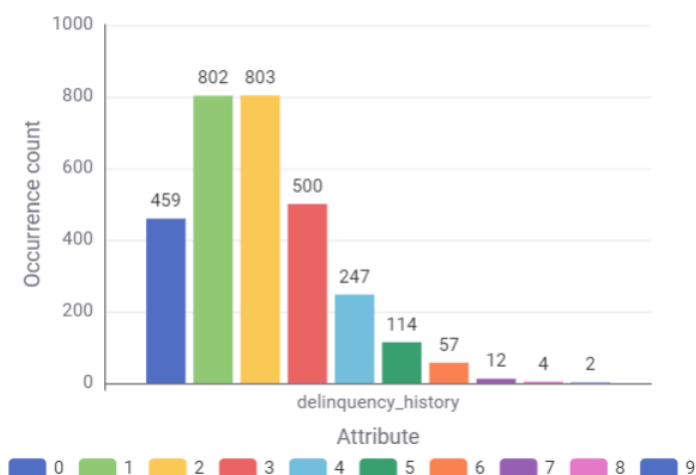


Current Balance

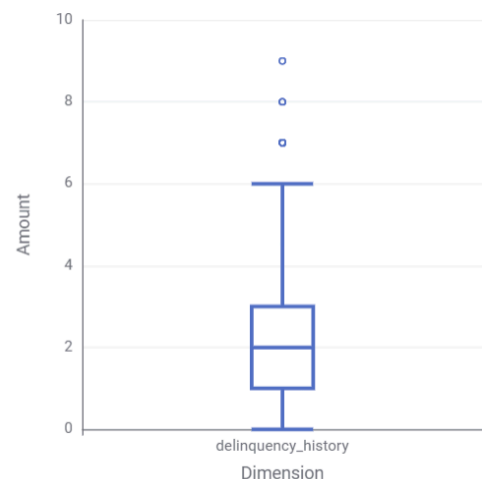


18. Delinquency history: Ratio, range 0 - 9. Distribution: Right-skewed (most borrowers have 1-3 delinquencies, with a long tail of repeat offenders). Key statistics: 68.8% of borrowers have 2 or fewer delinquencies 85.5% of borrowers have 3 or fewer delinquencies Only 6.3% of borrowers have 5 or more delinquencies Very few borrowers (0.6%) have 7+ delinquencies Box plot statistics: Min = 0, Q1 = 1, Median = 2, Q3 = 3, Max = 9. The interquartile range (IQR = 2) indicates relatively low spread, with most borrowers clustered around the median. Key insight: The majority of borrowers have a clean or near-clean delinquency history (0-2 events). The long tail extending to 9 delinquencies represents a small but notable high-risk segment (0.6% of borrowers).

Delinquency History

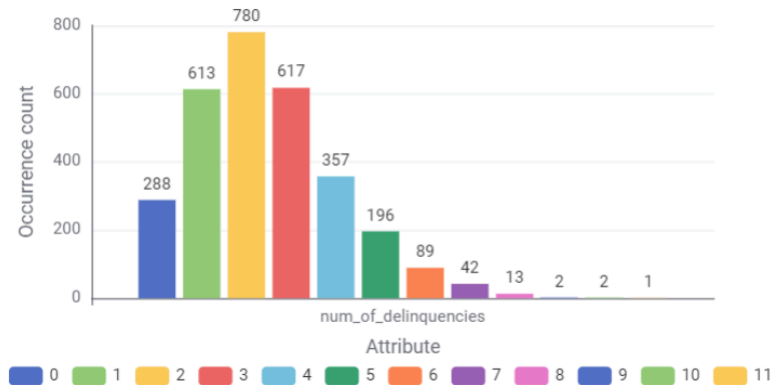


Delinquency History

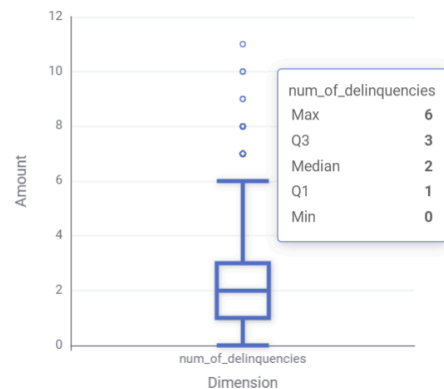


19. Number of delinquencies: Ratio, range 0 - 11. Distribution: Right-skewed (most borrowers have 1-3 delinquencies, with a long tail of repeat offenders). Key statistics: 56.0% of borrowers have 2 or fewer delinquencies 76.6% of borrowers have 3 or fewer delinquencies 88.5% of borrowers have 4 or fewer delinquencies Only 5.0% of borrowers have 5 or more delinquencies Very few borrowers (0.6%) have 8+ delinquencies Box plot statistics: Min = 0, Q1 = 1, Median = 2, Q3 = 3, Max = 11. The interquartile range (IQR = 2) indicates relatively low spread, with most borrowers clustered around the median. Comparison with Delinquency History (previous attribute): Delinquency History (total past events) had a similar distribution but slightly higher frequencies at lower counts Number of Delinquencies (current/recent) shows a similar pattern, confirming consistency in the data Key insight: The majority of borrowers have 0-3 current delinquencies (76.6%). The long tail extending to 11 delinquencies represents a small high-risk segment (1.2% of borrowers have 6+ delinquencies).

Number of Delinquencies

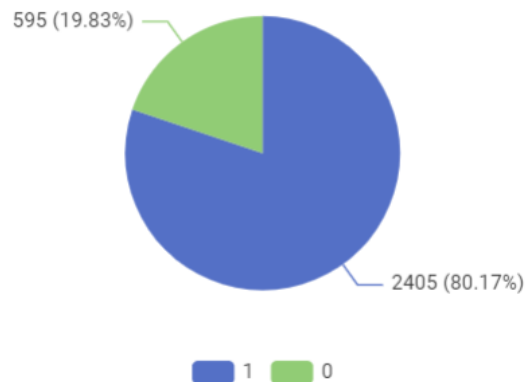


Number of Delinquencies



20. Loan paid back: Nominal (Binary). Distribution: The dataset is imbalanced, with the majority of loans being successfully repaid. Specifically, 2,405 borrowers (80.17%) repaid their loans, while 595 borrowers (19.83%) defaulted. Key insight: The default rate of approximately 20% is relatively high compared to traditional lending standards but may be typical for this borrower population, given the observed debt-to-income ratios and delinquency histories. Implications for analysis: When

Loan Paid Back



building predictive models (e.g., to identify borrowers at risk of default), techniques such as class weighting or resampling may be necessary to address the class imbalance between repaid (80%) and defaulted (20%) loans.

1A. 3 – Exploration

Explored relationships:

Category 1: Loan Outcome vs. Risk Factors

To understand what distinguishes borrowers who default from those who repay, box plots were used to compare the distributions of key continuous risk factors across the two loan outcome groups. These comparisons include credit score, debt-to-income ratio, interest rate, loan amount, annual income, installment, current balance, and number of delinquencies. If defaulters exhibit systematically lower credit scores, higher debt-to-income ratios, higher interest rates, or lower incomes, these variables would be validated as effective predictors of default risk. Conversely, little or no difference would suggest that other factors, such as loan purpose or employment status, may be more influential.

Below are the statistical tables of loan repayment vs. credit score, debt-to-income-ratio, interest rate, loan amount, annual income and number of delinquencies.

Loan Repaid vs. Credit Score

Rows: 2 | Columns: 7

Name	Minimum	Maximum	25% Quantile	50% Quantile (Medi...	75% Quantile	Mean
No	464	812	486	649	812	646.571
Yes	440	850	499	685	850	670.429

Loan Repaid vs. Debt-to-income-ratio

Name	Minimum	Maximum	25% Quantile	50% Quantile (Medi...	75% Quantile	Mean
No	0.021	0.627	0.021	0.212	0.52	0.262
Yes	0.01	0.581	0.01	0.144	0.424	0.211

Loan Repaid vs. Annual Income

Name	Minimum	Maximum	25% Quantile	50% Quantile (Medi...	75% Quantile	Mean
No	6,000	226,065.07	6,000	35,437.04	102,915.9	65,258.749
Yes	6,000	367,316.11	6,000	37,158.6	102,200.25	85,530.916

Loan Repaid vs. Loan Amount

Name	Minimum	Maximum	25% Quantile	50% Quantile (Medi...	75% Quantile	Mean
No	500	41,532.24	500	14,809.44	38,483.69	17,899.72
Yes	500	47,865.22	500	14,531.48	38,469.8	18,724.571

Loan Repaid vs. Interest Rate

Name	Minimum	Maximum	25% Quantile	50% Quantile (Medi...	75% Quantile	Mean
No	6.04	19.62	6.87	13.11	19.3	13.037
Yes	3.2	22.51	5.74	12.3	18.65	12.411

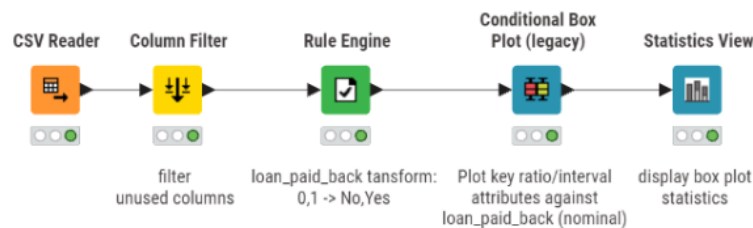
Loan Repaid vs. Number of Delinquencies

Name	Minimum	Maximum	25% Quantile	50% Quantile (Medi...	75% Quantile	Mean
No	0	10	0	2	7	3.571
Yes	0	11	0	2	6	3.286

Attribute compared against loan repayment	Findings
Credit score	Key Difference: Clear separation. Borrowers who repaid ("Yes") have a higher median (685 vs 649) and a higher mean (670 vs 647). Finding: Higher credit score is strongly associated with loan repayment.
Debt-to-income ratio	Key Difference: Clear separation. Repayers have a lower median (0.144 vs 0.212) and a lower mean (0.211 vs 0.262). Finding: Lower DTI (less existing debt relative to income) is associated with repayment.
Annual income	Key Difference: Large difference in means (Repayers: \$85,531 vs Non-repayers: \$65,259) but medians are close (\$37,159 vs \$35,437). Finding: The mean is pulled up by a subset of very high-income repayers (max \$367k vs \$226k). Income matters, but the effect is not uniform across all repayers.
Loan amount	Key Difference: Almost no difference. Medians (\$14,531 vs \$14,809) and means (\$18,725 vs \$17,900) are very close. Finding: Loan amount

	alone does NOT predict repayment.
Interest rate	Key Difference: Moderate separation. Repayers have a lower median (12.30% vs 13.11%) and a lower mean (12.41% vs 13.04%). Finding: Lower interest rates (likely due to better credit) are associated with repayment.
Number of delinquencies	Key Difference: Small but consistent. Repayers have a lower mean (3.29 vs 3.57) and a lower 75th percentile (6 vs 7). Finding: Repayers have slightly fewer past delinquencies, but the difference is not as dramatic as with credit score or DTI.

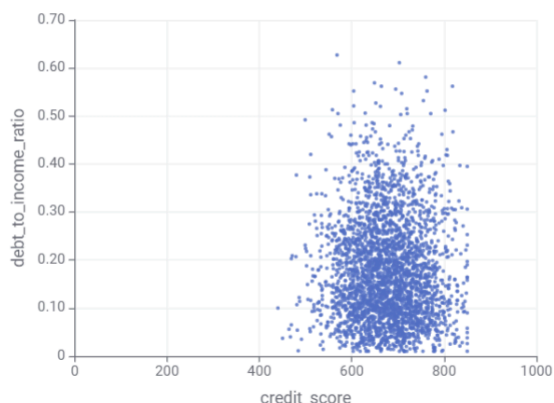
Takeaway: Borrowers who repay their loans have higher credit scores, lower debt-to-income ratios, and slightly lower interest rates — but the loan amount itself does not distinguish repayers from defaulters.



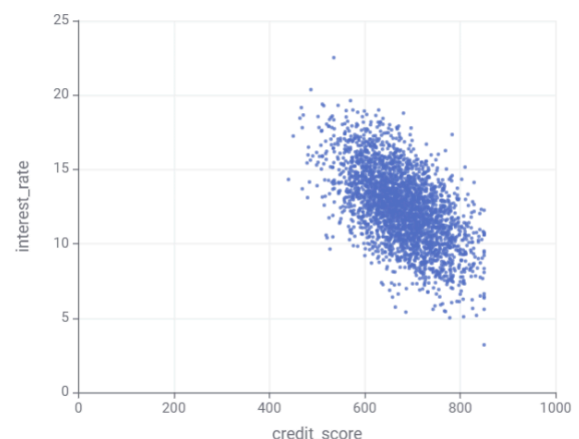
Category 2: Borrower Risk Profile Relationships

Scatter plots were used to examine pairwise relationships among borrower risk indicators, including credit score, debt-to-income ratio, interest rate, annual income, and delinquency history. These explorations aim to identify whether intuitive relationships hold in the data, for example, whether higher credit scores correspond to lower interest rates and lower debt-to-income ratios. Strong correlations in expected directions would confirm the internal consistency of the dataset, while unexpected or weak correlations might indicate unique borrower behavior or data anomalies worth investigating further.

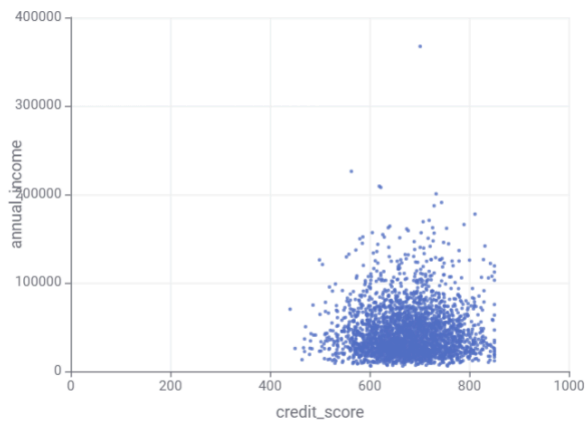
Credit Score vs. Debt-to-income Ratio



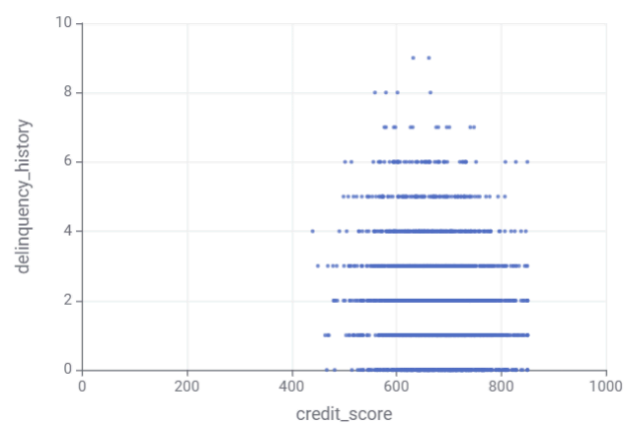
Credit Score vs. Interest Rate



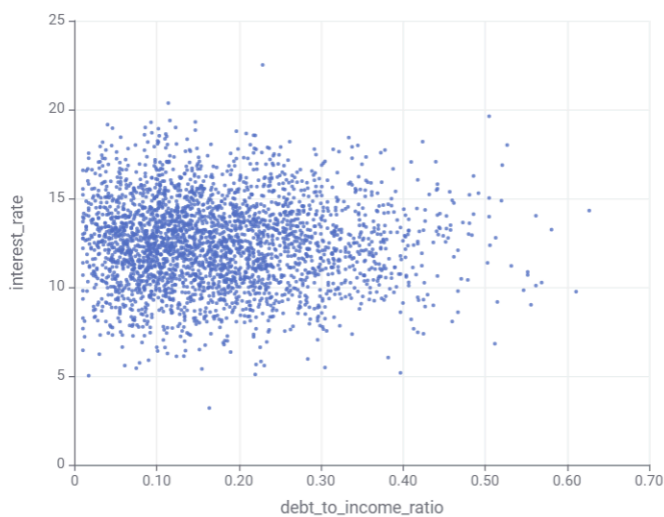
Credit Score vs. Annual Income



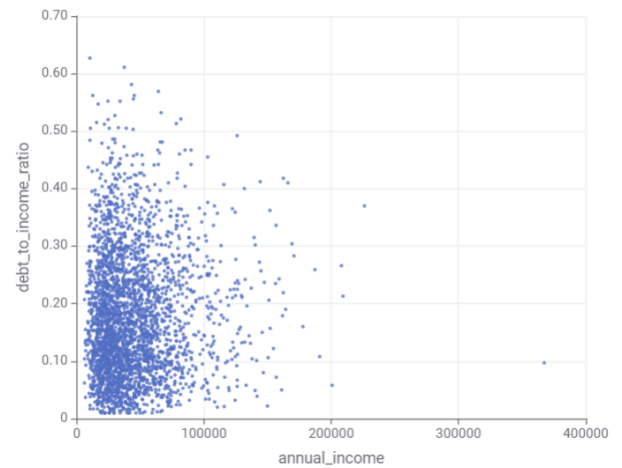
Credit Score vs. Delinquency History



Debt-to-income Ratio vs. Interest Rate



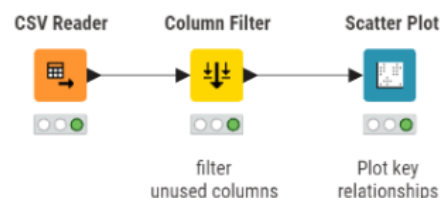
Annual Income vs. Debt-to-income Ratio



Relationships explored	Findings
Credit score vs. debt-to-income ratio	Key Difference: Moderate negative correlation. Borrowers with credit scores above 700 have DTI concentrated below 0.4, while those below 600 show DTI between 0.5-0.8. Finding: Higher credit scores are generally associated with lower debt burdens, though exceptions exist.
Credit score vs. interest rate	Key Difference: Strong negative correlation. Interest rates decrease from 15-20% (credit score <580) to 5-8% (score >740). Finding: Credit score is the strongest predictor of interest rate, validating proper lender risk pricing.
Credit score vs. annual income	Key Difference: No correlation. High-income earners (\$150k+) span

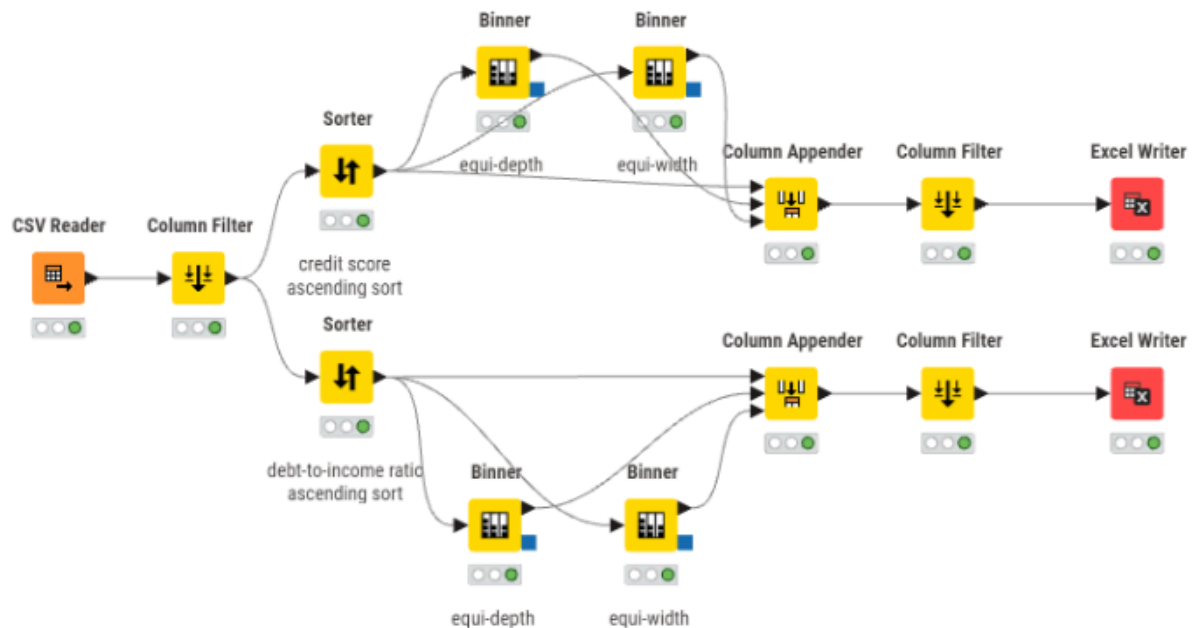
	significant credit score range (580-800+), and excellent credit (eg. 740+) appears across all income levels (\$30k-\$300k+). Finding: Income alone does not determine creditworthiness, payment behavior matters more.
Credit score vs. delinquency history	Key Difference: Moderate negative relationship. Borrowers with 2+ past delinquencies are heavily concentrated in credit score ranges below 650. Borrowers with 0-1 delinquencies dominate scores above 700. Finding: Delinquency history is a primary driver of credit score.
Debt-to-income ratio vs. interest rate	Key Difference: Weak to moderate positive correlation. DTI <0.3 yields rates 5-12%, DTI 0.3-0.4 yields rates 8-15%, DTI >0.4 yields rates 10-18%. Finding: Higher DTI increases interest rates, but the relationship is weaker than credit score's impact.
Annual income vs. debt-to-income ratio	Key Difference: No correlation. Low and high incomes both show DTI across the full 0.1-0.8 range. Finding: High earners can have high DTI; low earners can have low DTI.

Takeaway: Credit score is the strongest predictor of both interest rates and loan repayment risk, with a clear inverse relationship. Delinquency history heavily drives credit scores while debt-to-income ratio shows moderate correlation with credit scores and weak correlation with interest rates. Notably, annual income shows no correlation with either credit score or DTI, confirming that creditworthiness is determined by payment behavior and debt management, not by how much someone earns. Loan amount (from Category 1) already showed no predictive power for repayment, and these scatter plots reinforce that risk profiling must rely on credit history and DTI, not income or loan size alone.



Task 1B. Data Preprocessing

1B. 1 – Binning

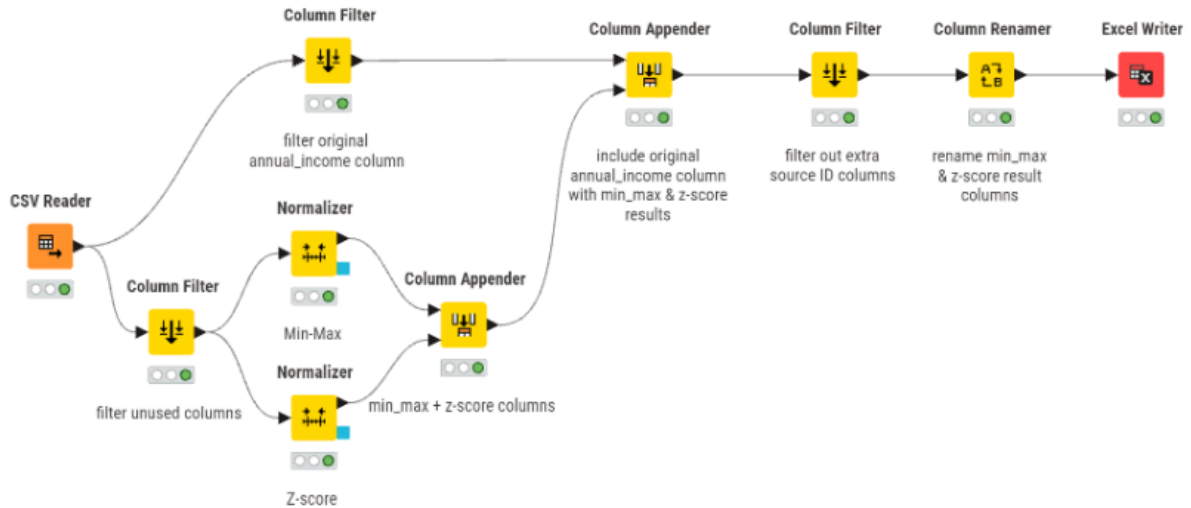


For equi-depth binning for both credit score and debt-to, 25 bins provide statistical significance while maintaining granularity for smoothing. Auto-binner with 25 equi-depth (frequency) bins, each containing 120 values each.

For equi-width binning, both credit score and debt-to-income ratio use auto-binner with 20 equi-width bins to divide range into balanced slices as 20~30 bins provide the best balance for smoothing and statistical significance.

Column appenders and filters are used separately to write in individual files to preserve order of sorted data.

1B. 2 – Normalisation

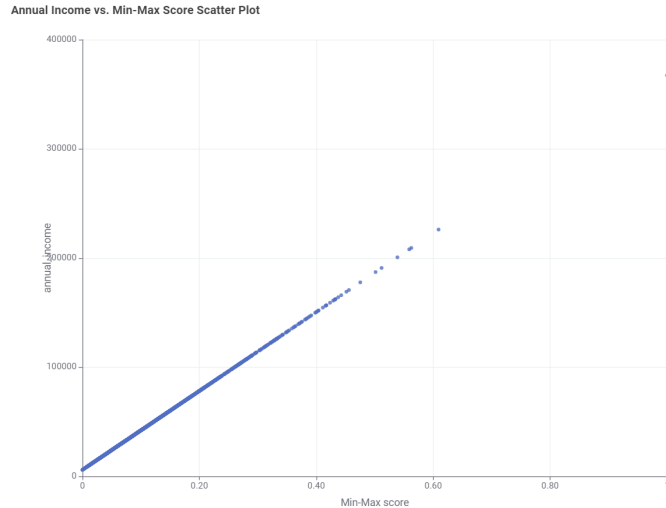


- I. Min-max technique performs a linear transformation over the original data. Min-max normalisation preserves the relationships between the original data values. Suppose that $\min_K$ and $\max_K$ are the minimum and maximum values of an attribute K . Min-max normalisation maps a value of K , indicated by v_n , to a new value v'_n within the range $[\text{new min}_K, \text{new max}_K]$ by calculating:

$$v'_n = \frac{v_n - \min_K}{\max_K - \min_K} (\text{new max}_K - \text{new min}_K) + \text{new min}_K$$

In our case, we transform the annual_income values onto the range $[0.0 - 1.0]$. From the data set, the minimum and maximum annual incomes ($\min_K$ and $\max_K$) are 6,000 and 367,316.11.

So, the min-max value for attribute K (annual_income) are obtained by using:



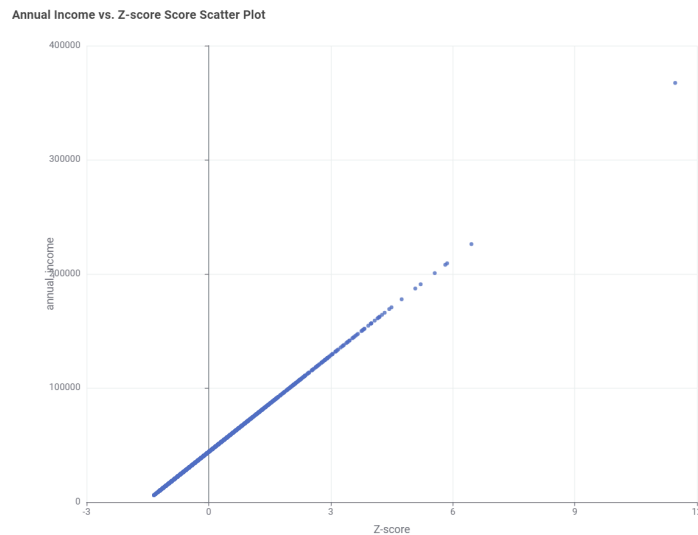
$$\frac{K - 6,000}{367,316.11 - 6,000} (1.0 - 0) + 0.0 = \frac{K - 6,000}{361,316.11}$$

- II. Z-score (also known as zero-mean) normalisation normalises attribute values using the average (i.e., mean) and standard deviation of K . Let v_n be the value of K . v_n is normalised to v'_n using the formula:

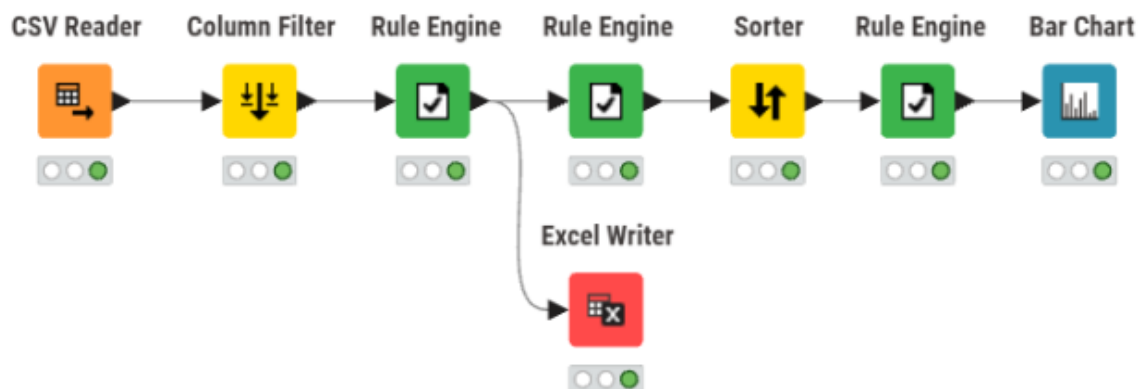
$$v'_n = \frac{v_n - \mu}{\sigma_K}$$

Here, $\mu = \frac{1}{t} (v_1 + v_2 + \dots + v_t)$

represents the mean of attribute K , and σ_k is the standard deviation, or the square root of the variance of K . Z-score normalisation is useful when there are outliers that dominate the min-max normalisation, or when the actual minimum and maximum of attribute K are unknown. In our dataset, we have absolute minimum and maximum annual salaries along with outliers.



1B. 3 – Age Categorisation



Categorisation and plot:

1. Read data file.
2. Filter columns to keep source ID and age columns.
3. Categories age into required groups:

Age < 22	=>	Young
Age > 21 & < 35	=>	Early Adulthood
Age > 34 & < 45	=>	Early Middle Age
Age > 44 & < 65	=>	Late Middle Age
Age > 64	=>	Late Adulthood

Expression	
☒	1 <code>\$age\$ < 22 => "young"</code>
☒	2 <code>\$age\$ >21 AND \$age\$ <35 => "early adulthood"</code>
☒	3 <code>\$age\$ >34 AND \$age\$ <45 => "early middle age"</code>
☒	4 <code>\$age\$ >44 AND \$age\$ <65 => "late middle age"</code>
☒	5 <code>\$age\$ >64 => "late adulthood"</code>
	6
	7
	8

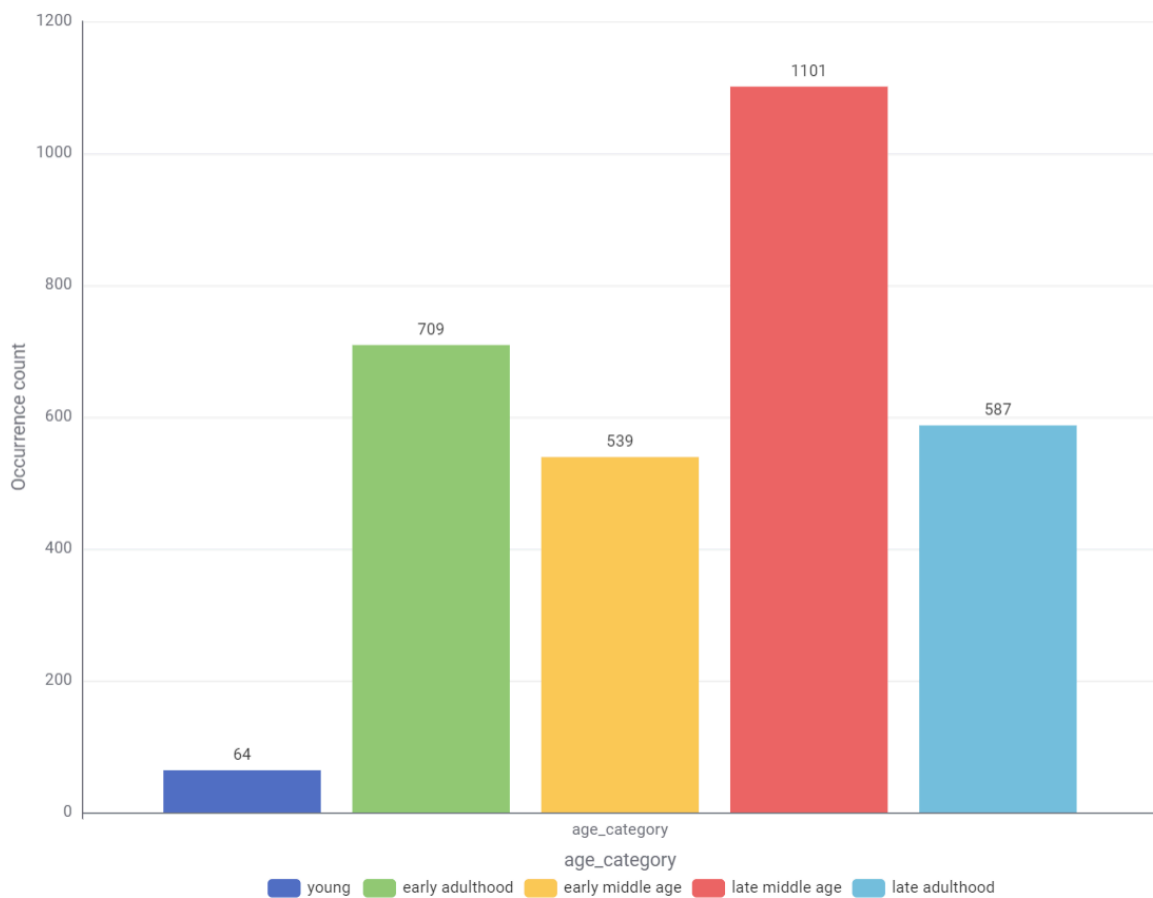
☒ Append Column: ☒

☐ Replace Column: ☐

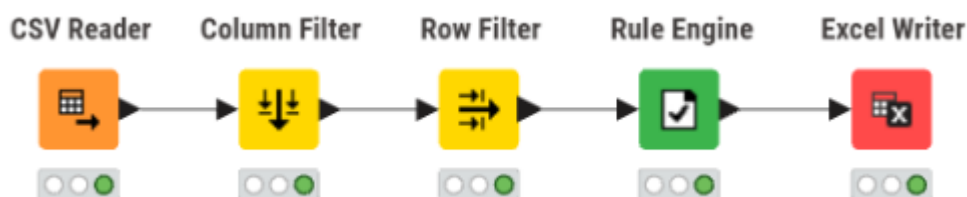
4. Assigning integers to age categories: young = 1, early adulthood = 2.. etc.
5. Sorting (integer) categories in ascending order.
6. Changing back integers to category labels: 1 = young, 2 = early adulthood.. etc.
7. Bar chart of categorical frequency.
8. Write the categorised ages data to excel file.

Steps 4 and 6 (corresponding to node 4 and 6) help arrange the categories correctly, in ascending age order (young, early adulthood...) for the plot, which otherwise would be arranged lexically.

Age Category Frequency Bar Chart



1B. 4 – Gender Binarisation



Gender binarisation process:

1. CSV read
2. Filter columns to keep source ID and gender columns
3. Filter out rows containing “Other” gender
4. Binarise: Male = 0, Female = 1; assign results to new “gender_binary” column
5. Write results in excel file

Reasoning:

The data file initially contains 3 gender types: Male, Female, and Other. Filtering out “Other” gender rows reveals that 2932 of 3000 total rows consist of Male and Female genders, contributing to 97.7% of total genders. Only 68 rows, 2.3% of the rows are classified as “Other” gender. Removal of these 68 (2.3%) rows is below the typical 5-10% threshold for acceptable data loss. This simplifies the binarisation process to 2 genders without having to classify the third gender along with any of the others as the removal of 68 records slightly reduces sample size but does not materially affect the statistical power or generalisability of the analysis. Source ID is retained to match up the gender and gender_binary columns.

The image shows a data processing workflow and a filter configuration panel.

Workflow: CSV Reader → Column Filter → Row Filter → Rule Engine → Excel Writer. The Row Filter step is highlighted with a blue box and labeled "Add comment".

Filter Configuration Panel:

- Criterion 1:** Filter column: gender, Operator: Is not equal.
- Case matching:** Case insensitive (selected).
- Value:** other.
- Buttons:** Discard, Apply and Execute.

Data Table:

Rows: 2932 | Columns: 2

#	RowID	source ID	gender
		Number (Integer)	String

Summary of Findings

1. Credit score is the single strongest predictor of loan repayment: Borrowers who repaid their loans had a median credit score of 685 compared to 649 for defaulters — a clear and actionable separation. Credit score also showed a strong negative correlation with interest rate (higher score → lower rate), confirming that lenders are pricing risk correctly.

Business implication: Credit score should be the primary feature in any default risk model.

2. Debt-to-income ratio (DTI) is the second most important risk indicator: Repayers had a median DTI of 0.144 (14.4% of income to debt) versus 0.212 (21.2%) for defaulters. The

DTI distribution is right-skewed, with most borrowers clustered below 0.30 and a long tail extending to 0.627 (high-risk segment).

Business implication: Borrowers with DTI above 0.40 warrant additional scrutiny or higher rates.

3. Annual income is a misleading risk factor when used alone: While the mean income of repayers (\$85,531) is much higher than defaulters (\$65,259), the medians are nearly identical (\$37,159 vs \$35,437). This indicates that a small number of very high-income repayers pull the mean upward. Scatter plots confirmed no correlation between income and credit score or between income and DTI.

Business implication: Income should not be used as a standalone risk filter. A high earner can have poor credit, and a low earner can be creditworthy.

4. Loan amount does not distinguish repayers from defaulters: Medians (\$14,531 vs \$14,809) and means (\$18,725 vs \$17,900) are almost identical across repayment outcomes. Loan size alone carries no predictive power.

Business implication: Not to use loan amount as a risk variable. Focus on credit score and DTI instead.

5. Age is uniformly distributed with no outliers: Borrowers range from 21 to 75 years, with each 5-year bin containing 233–288 customers (mean $\approx$ 260). No age group dominates, and no age-based outliers exist.

Business implication: Age does not introduce sampling bias. Analysis across age groups is balanced and comparable.

6. Education level shows an ordinal risk gradient: Bachelor's degree is most common (41.4%), followed by High School (28.9%). Higher education (Master's, PhD) correlates with higher income, though income itself is not a direct risk predictor.

Business implication: Education may serve as a useful proxy for risk when credit history is unavailable, but should not replace credit score.

7. Two distinct borrower segments emerge from current balance distribution: Current balance is bimodal: one peak in the low range (\$648–\$50,000) and another in the medium range (\$50,000–\$100,000). This suggests two borrower types: those who pay down balances regularly, and those who carry significant debt.

Business implication: Investigate these two segments separately in future analysis. They may have different default behaviours.

8. Delinquency history drives credit score: Borrowers with 2+ past delinquencies are concentrated in credit score ranges below 650. Borrowers with 0–1 delinquencies dominate scores above 700. This confirms that payment behaviour — not income or loan size — determines creditworthiness.

Business implication: Delinquency history should be included alongside credit score in risk models.

9. Default rate is approximately 20%: Of 3,000 borrowers, 2,405 (80.17%) repaid their loans, while 595 (19.83%) defaulted. This is relatively high compared to traditional lending standards but appears consistent with the borrower population's DTI and delinquency profiles.

Business implication: The 20% default rate suggests either a high-risk portfolio or a need to tighten underwriting criteria — specifically around DTI and credit score thresholds.

10. Recommended clusters for formal investigation: Visual exploration (scatter plots and histograms) revealed natural groupings that warrant formal clustering analysis:

Cluster	Defining characteristics
Prime borrowers	Credit score > 700, DTI < 0.25, low delinquencies
Near-prime borrowers	Credit score 620–700, DTI 0.25–0.40
Subprime borrowers	Credit score < 580, DTI > 0.40, multiple delinquencies

Business implication: Formalise these three risk tiers to segment underwriting, pricing, and collection strategies.